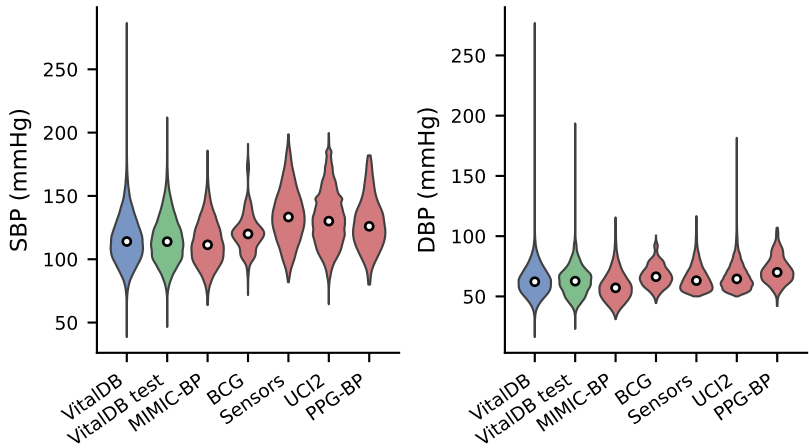


a

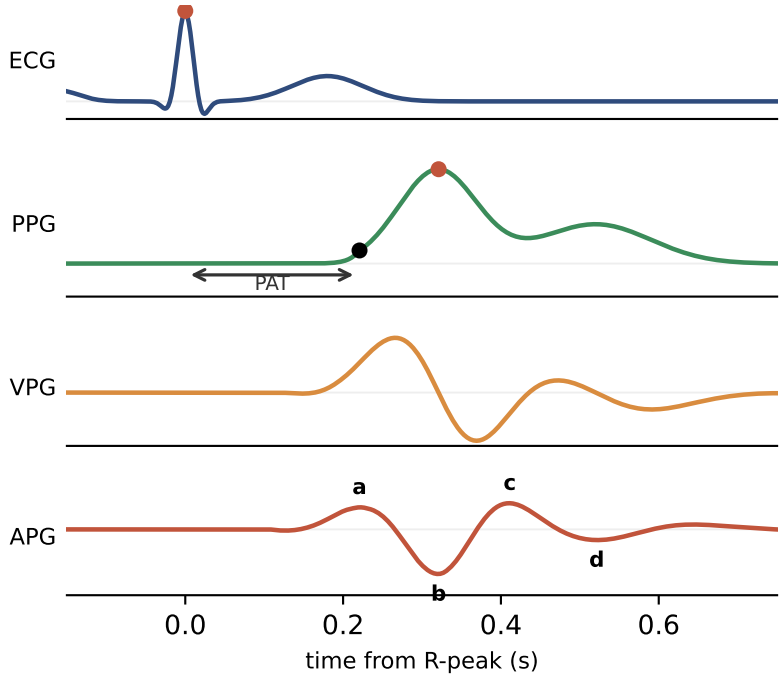


b

Datasets

Dataset	Role	Seg	Subj	SBP/DBP	Age [5-95]	Chan
VitalDB	train	465k	1,293	115/63	61 [31-79]	ECG+PPG
VitalDB test	ID	58k	144	115/63	60 [30-79]	ECG+PPG
MIMIC-BP	OOD	137k	1,524	113/58	--	ECG+PPG
BCG	OOD	3k	40	121/67	28 [19-65]	PPG
Sensors	OOD	11k	1,195	134/65	57 [33-81]	PPG
UCI2	OOD	411k	10,793	132/67	--	PPG
PPG-BP	OOD	619	218	128/72	58 [25-80]	PPG

c Feature extraction (ECG, PPG, 1st/2nd deriv.)

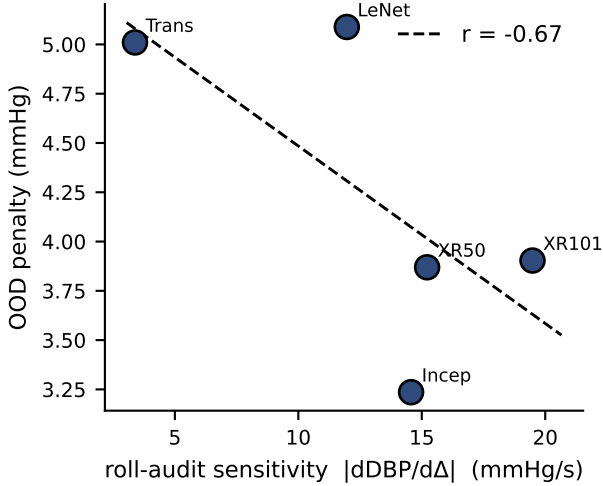


d

DBP MAE (mmHg); audit slope, all CIs span zero

Model	size	ID	OOD	audit slope
LeNet1d	659k	8.5	13.5	-0.009
Inception1d	472k	8.6	11.8	-0.003
XResNet50	887k	8.7	12.6	-0.008
XResNet101	1.8M	8.6	12.5	-0.005
Transformer	107k	8.6	13.6	-0.004
LightGBM	~8k	8.2	15.1	n/a
LightGBM (83)	50k lv	8.10	14.65	n/a
LightGBM +demo	50k lv	8.28	16.76	n/a
LightGBM 1 tree	32 lv	8.84	13.13	n/a
LightGBM PAT only	6 feat	9.31	11.09	n/a

e Mechanism vs OOD robustness



f

Decodability is flat across models (number = |roll slope|, mmHg/s)

